

Which Pair Is It About?

Query-Conditioned Verification in a Deployed Biotic-Interaction Extraction Pipeline

Anonymous ACL submission

Abstract

Literature-mining pipelines that populate biotic-interaction databases propose candidate (taxon, relation, taxon) triples with a supporting passage, then filter them with a sentence-level interaction classifier before curation. We audit that filter on 97 expert-graded candidates from a deployed system. The filter is close to inert by construction: a rule layer runs first, so every one of the 175,588 candidates in the retrieval pool already carries two taxon surface forms and an interaction surface form. It accepts 89 of 97 and reaches precision 0.843, not significantly above the 0.773 obtained by accepting everything (one-sided exact binomial $p = 0.071$). We decompose its false positives by mechanism: pair-binding failures, in which every component is correct but the relation is not asserted between *this* pair; predicate misbindings; and entity failures. Conditioning the same decision on the candidate triple addresses the first class directly. On the full 437-item benchmark, with the decision threshold fitted on two source blocks and applied to the third so that no row is scored by a threshold fitted on it, a distilled 110M cross-encoder reading the triple as a query raises F1 from 0.816 to 0.860 — 62 items fixed against 47 broken, which is *not* significant (McNemar $p = 0.180$). The result that does separate the two is recall-side: on 41 curated pipeline rejections the deployed filter scores F1 0.000 by construction and the verifier reaches 0.700. We state the limit plainly: the contribution is a calibrated, tunable score that reaches a class of true positives the deployed filter cannot reach at all, not a significant gain in verdict accuracy. Entity disambiguation and argument order survive the reformulation; we make no direction claim.

1 Introduction

Biotic-interaction databases such as GloBI (Poe-len et al., 2014) are populated in part by text mining. A retrieval system proposes candidate triples

(s_1, r, s_2) together with the passage that allegedly supports them, and a filter decides which candidates are worth a curator’s attention. That filter is almost always a *sentence-level* interaction classifier: it is shown the passage and asked whether it describes a biotic interaction.

On 97 expert-graded candidates from a deployed system, that filter accepts 89. Its precision, 0.843, is higher than the 0.773 obtained by accepting every candidate, but not significantly so (one-sided exact binomial $p = 0.071$); its recall is 1.000. This is not a tuning accident, and it is not carelessness by the people who built it. A rule layer runs upstream of the classifier, and every candidate it emits already contains two taxon surface forms and an interaction surface form — on all 175,588 rows of the retrieval pool. The property the classifier was trained to detect is therefore, as a lexical matter, constant across its entire deployment distribution. It was trained where negatives lack interaction language; at deployment no negative lacks it.

What the pipeline actually needs to know is narrower: not whether the passage describes an interaction, but whether it asserts *this* relation between *these two* taxa. We decompose the filter’s 26 false positives along that distinction and find three mechanisms (§4): 11 pair-binding failures, where every component is individually correct and only the binding is wrong; 8 predicate misbindings, where the interaction term’s real arguments are other tokens; and 7 entity failures. The first class is invisible to any checker that validates s_1, r and s_2 separately, which bounds every component-wise architecture at precision 0.802 at full recall.

Conditioning the decision on the candidate triple addresses the pair-binding class directly. A 110M cross-encoder, distilled from a teacher that was itself asked the triple-specific question, raises F1 on the full 437-item benchmark from 0.816 to 0.860 (§6) and recovers true positives the pipeline had already discarded. We are explicit that this difference

is not significant: 62 items fixed against 47 broken, McNemar $p = 0.180$. Removing the query from its input at inference collapses its ranking to near the base rate, which is the controlled evidence that the reformulation rather than the encoder is doing the work.

We also state the ceiling, because a reviewer will find it otherwise. At the deployed filter’s own acceptance level the verifier reproduces its confusion matrix: with 75 supported pairs among 97, any system accepting 89 incurs at least 14 false positives, and both attain that floor. Nor does the verifier win everywhere. Block by block it is better on TEST299 ($0.787 \rightarrow 0.859$) and far better on REJECT50 ($0.000 \rightarrow 0.700$), and *worse* on BIOTX100 ($0.915 \rightarrow 0.884$)—the block the deployed filter was in effect tuned against. The contribution is therefore not superiority at the deployed operating point, which is unavailable to anyone, but the existence of an operating curve where a fixed verdict stood.

Finally, scope: of 50 curated pipeline rejections only 11 are classifier errors, the rest failing upstream at the rule layer or at taxon resolution. A classifier contribution addresses a minority of this pipeline’s rejection volume, and we claim no more.

Contributions.

- An audit of a deployed filter against its own expert-graded output, with the structural reason it accepts almost everything (§2–§3).
- A mechanism decomposition of its false positives, a bound on component-wise remedies, and a measurement of what the reformulation repairs (§4, §6).
- An operating-point analysis reported against our own interest, and a recall-side benchmark with its leakage audited (§6.3, §6.4).
- Four negative results and a measurement of why argument order is not currently evaluable (§7, §A).

2 Setting

The pipeline under audit. BiotXplorer (Ruch et al., 2024) proposes candidate biotic-interaction triples from the literature indexed by SIB Literature Services (Gobeill et al., 2020). For each passage it matches taxon surface forms against a taxonomy and interaction surface forms against a vocabulary derived from the Relation Ontology, and emits every resulting (s_1, r, s_2) combination together with

the passage, the three matched surface strings, their canonical forms, and heuristic retrieval scores. Candidates are therefore proposed by *co-occurrence*: a passage naming two taxa near an interaction term produces a candidate whether or not it asserts that interaction between those two. A filter then decides what reaches a curator, and in the deployed system that filter is a sentence-level interaction classifier.

Why the classifier’s feature is inert at this stage.

The rule layer runs before the classifier, and it is also what defines the pool. We pulled the deployment pool in full: 175,588 candidate rows over 165,325 distinct passages and 115,404 distinct triplet keys. Both taxon surface forms and the interaction surface form are non-empty on 175,588 of 175,588 rows, because the presence of an interaction term is the pool’s construction filter; an independent lexical scan for GloBI interaction vocabulary fires on 99.82% of the same passages. Every input the classifier sees at deployment therefore satisfies, lexically, the property it was trained to discriminate. The generalisation is a partial-input failure whose cue was installed by the pipeline’s own architecture rather than by an annotator: the missing input is the query the pipeline itself issued.

Benchmarks. One benchmark of 437 rows, 246 positive, in three blocks of different provenance, and we report every block rather than pooling them into a single number that hides the disagreement between them.

BIOTX100 is retrieved candidates graded by a domain expert on four axes: whether each taxon surface form was correctly resolved (92/100 and 98/100), whether the relation was correct (90/100), and whether the whole triple is supported by its passage (65/100). After removing the three training duplicates it contributes 97 rows, 75 of them positive at the species level. It uniquely records the deployed filter’s own accept/reject decision, which is what makes §3 possible. Every row has rank 1, so this is precision at one retrieved candidate. REJECT50 is curated pipeline *rejections*, 41 clean rows with 9 true positives wrongly discarded; it is the only recall-side instrument in this paper, and the deployed filter scores F1 0.000 on it by construction. TEST299 is 299 pair-level items from expert curation, 0.542 positive.

The three blocks do not carry the same gold. BIOTX100 is the only block graded at the level of the whole triple; TEST299 was curated as a pair-level set and its stored relation term was at-

tached by the retrieval pipeline rather than by an annotator. We therefore evaluate everything at the level all three blocks share—does this pair interact in this passage—and treat triple-level grading as available on BIOTX100 alone. That choice costs us the sharper claim and is the reason the headline is a pair-level result.

A query-representation correction. Each candidate carries both a canonical string (`species1_term`) and the surface form matched in the passage (`species1_form`). These differ on 34, 38 and 59 of 100 rows for s_1 , s_2 and r , and the canonical string is *absent from the passage* on 31, 27 and 54 rows. Scoring a verifier with canonical strings asks it about text it cannot see. An earlier version of this work did exactly that; every number reported here uses the surface-form query, the representation the model consumes at inference, and comparisons to our own prior reports are void.

Evaluation protocol. Labels are species-level throughout—*do these two taxa interact in this passage*—which is the semantics the benchmark was curated under and what the deployed filter is for. The whole passage is the second segment for every row; one row of 437 exceeds the 256-wordpiece window.

Thresholds are the place this evaluation is easiest to get wrong, so we state the policy explicitly and report three. **Pre-specified** is $\tau = 0.5$, fixed in advance. **Block-held-out** fits τ on two of the three source blocks and applies it to the third, so every row is scored by a threshold that never saw it, and every row is used. This is the policy we report. **Oracle** maximises F1 on the reporting set; it appears only as a labelled upper bound. An earlier version of this work reported an operating point derived from a prior-shift rule applied to the reporting set’s own gold labels. That is a threshold fitted on the data it scores, every number computed under it is withdrawn, and the present numbers share no threshold with it.

All comparisons report the accept-everything baseline, and intervals are 20,000-sample bootstraps or Clopper–Pearson. Where three seeds exist we give the per-seed mean and standard deviation; where we also report the average of the three checkpoints’ probabilities we call it an ensemble, because it is one and it does not behave like a single model.

On contamination: the benchmark ships an

System	TP	FP	FN	TN	P	F1
Accept everything	75	22	0	0	0.773	0.872
Deployed filter	75	14	0	8	0.843	0.915
Verifier @ 0.04	65	7	10	15	0.903	0.884

Table 1: BIOTX100 after removing the three training duplicates: 97 candidates, 75 with an interacting pair. The deployed filter’s precision is not significantly above accepting everything (one-sided exact binomial $p = 0.071$), which is the audit’s point. On this block the verifier buys precision ($0.843 \rightarrow 0.903$) and pays recall for it, ending *below* the filter on F1; the difference is not significant either way (McNemar $p = 0.646$, 8 fixes against 11 breaks). τ is fitted on the other two blocks, never on this one. Verifier figures average three checkpoints’ probabilities, which is an ensemble; per-seed figures are in Table 5.

`in_train` flag on 9 rows, and a 5-gram Jaccard scan against all 48,338 training passages finds 3 more at Jaccard 1.000—exact duplicates, all of them in BIOTX100. All 12 are removed, which is why BIOTX100 contributes 97 rows here. An earlier version of this paper stated that BIOTX100 shares no passage with training data; that was wrong.

3 Auditing the deployed filter

Table 1 gives the central observation. The deployed filter accepts 91 of 100 candidates and reaches precision 0.843 against 0.773 for accepting everything, a difference this benchmark cannot resolve ($p = 0.119$; Clopper–Pearson intervals $[0.610, 0.804]$ and $[0.548, 0.743]$). Its recall is 1.000: it rejects nothing that should be kept.

Two things must be said precisely here, because the obvious summary — “the filter does not filter” — is stronger than the data support. First, the filter is not uninformative. Its accept set is not independent of whole-triple support: all nine of its rejections are correct, Fisher’s exact test on its 2×2 decision table gives $p = 3.7 \times 10^{-5}$, and against accepting everything its *accuracy* advantage is significant (exact McNemar on 9 discordant pairs, $p = 0.004$). Second, what it cannot do is convert that signal into precision, because it accepts 91 of 100 candidates and exposes no way to accept fewer.

The sharpest form of the observation conditions on the sentence-level question already being answered affirmatively. Among the 90 candidates whose relation the expert graded correct, the filter accepts 83, and of the 25 that nonetheless fail to support their triple it rejects 7: precision 0.783

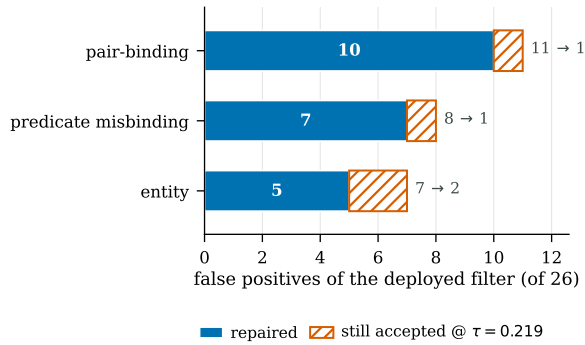


Figure 1: The deployed filter’s 26 triple-level false positives by mechanism, and what survives the verifier at an illustrative $\tau = 0.219$. This is a diagnostic of *what the failures are*, not an operating-point result: $\tau = 0.219$ is not the threshold reported anywhere else in this paper, and the per-row mechanism grades behind this figure are not in our release (see Limitations). The three per-class rates (10/11, 7/8, 5/7) are not distinguishable from one another at these cell sizes and we claim no differential.

against a subset base rate of 0.722, again not separable ($p = 0.131$). The verifier on the same 90 reaches 0.950 at 60 kept. Once the sentence-level question is settled, the filter has little discriminative power left over the question the pipeline actually asks.

The signal it does carry is also not the signal it was trained on. Its decisions reach a Matthews correlation of 0.128 with the sentence-level axis it was trained on — does this passage describe a biotic interaction — against 0.560 with whether both taxa were correctly resolved and 0.429 with whole-triple support. A classifier trained to detect interaction language, then deployed where interaction language is a constant (§2), ends up acting on incidental correlates of the question it is not being asked.

4 What the false positives are made of

This section is the one place we work at the level of the whole triple, because it is a diagnosis of *which component is wrong* and that question is meaningless without triple-level grades. BIOTX100 is the only block that has them. Everything reported in §6 is pair-level, on all three blocks.

The deployed filter’s 26 triple-level false positives are not homogeneous. We classify each by a fixed cascade over the expert’s three component grades: *entity* if either taxon grade is 0, else *predicate misbinding* if the relation grade is 0, else *pair-binding*. This yields 7 entity, 8 predicate and 11 pair-binding failures; Table 2 gives one worked

example of each. Applied to all 35 unsupported candidates rather than to the 26 the filter accepted, the same cascade gives 10, 9 and 16.

The two label levels disagree in a way that is itself informative. Scored at the pair level the same accepted set has only 15 false positives, decomposing as 8 pair-binding, 7 entity and *zero* predicate misbindings—by construction, since a misbound predicate between two taxa that do genuinely interact is still a correct pair-level positive. Eleven of the filter’s 26 triple-level errors are therefore invisible to the pair-level evaluation we report as our headline, which means that headline *understates* what triple conditioning buys. We prefer to understate it than to score two of the three blocks against a relation term no annotator ever checked.

The pair-binding class is the one that matters architecturally. Its members have both taxa correctly resolved *and* the correct relation type; the passage simply does not assert that relation between those two. Typically the relation’s true argument is a third organism named in the same sentence, or the sentence lists several taxa of which only a subset participate.

Two properties make this class hard for anything short of a joint decision: the relation is genuinely asserted, so relation checking cannot help, and both arguments are genuinely present and correctly resolved, so entity checking cannot either. What is wrong is a binding that exists only in the retrieval step’s combinatorics. The deployed filter is not blind to the class — 5 of its 9 correct rejections are pair-binding failures — it simply rejects too few candidates to matter.

A bound on component-wise verification. This bounds any component-wise architecture at every operating point. A verifier that checks s_1 , r and s_2 separately — however good each check is — rejects at most the 19 component-detectable errors among the 35, and has no signal whatever that separates the 16 pair-binding failures from supported triples. Accepting a fraction r of the 65 supported triples it therefore also accepts all 16, giving precision $65r/(65r + 16)$: 0.802 at full recall and 0.781 at $r = 0.877$. The deployed filter sits 0.088 *below* that bound at full recall, so it is not merely failing to exploit joint information — it is losing information the components alone would supply. Figure 2 plots the bound.

Mechanism (n)	Candidate triple	Passage (abridged)
Pair-binding (11)	(<i>M. marinum</i> , pathogen of, <i>M. tuberculosis</i>)	“ <i>Mycobacterium marinum</i> is a waterborne pathogen of fish and other ectotherms , closely related to <i>Mycobacterium tuberculosis</i> .”
Predicate misbinding (8)	(<i>Salmonella</i> Poona, affected by, cantaloupe)	“Attachment of <i>Salmonella</i> Poona to cantaloupe rind ... as affected by temperature of fruit and inoculum .”
Entity (7)	(<i>P. melanarius</i> , predator, Illiger)	“... the carabid predator <i>Pterostichus melanarius</i> (Illiger) ...”

Table 2: One graded example of each mechanism, from the deployed filter’s 26 false positives. The relation’s true object is italicised where it is not the queried taxon; in the entity case the taxonomic authority’s surname was matched as a taxon.

5 Conditioning the decision on the triple

The reformulation. Putting the pair in the input is not by itself the mechanism; making the *label move when the pair moves* is. In earlier work on this pipeline we added the taxon pair to a sentence-level classifier as a genuine segment pair, changing nothing else, and AUPRC did not move. The reason is that the question the model was *asked* changed while the question its labels *answered* did not: with sentence-level targets, two training items sharing a passage but differing in pair carry the same label, so gradient descent has no reason to attend to the query.

The teacher here is asked, per candidate, whether the passage supports *this* triple, so its labels vary across pairs on the same passage. It is a locally hosted 32B open-weight instruction model (Qwen Team, 2025), prompted zero-shot at temperature 0. On the BIOTX100 items for which we hold human gold it reaches precision 0.919 and F1 0.898, and against 150 human-graded items its overall mislabel rate is $22/150 = 14.7\%$, Clopper–Pearson [9.4%, 21.4%], split 10 false positives and 12 false negatives.

The corpora. VERIFIER-BASE is distilled from 20,000 teacher-labelled rows (0.289 positive). VERIFIER-PC adds three blocks for a total of 48,338 rows (0.343 positive): 16,546 *pair-contrast* rows, 6,201 actively sampled rows, and 5,591 rows in which the candidate’s argument order is reversed. The pair-contrast block is the point of the design: it contributes 2,017 passages that carry *different labels for different pairs*, of which VERIFIER-BASE’s 19,999 distinct passages contain exactly zero. All rows are drawn from the same rule-layer candidate pool as deployment, so 100% carry two taxa and an interaction form and the train/deploy mismatch that produced the original failure is closed by con-

τ	keep	TP	FP	P	R	F1	fix/brk	p
0.01	290	226	64	0.779	0.919	0.843	60/57	0.853
0.02	272	224	48	0.824	0.911	0.865	62/45	0.122
0.04	260	219	41	0.842	0.890	0.866	61/42	0.076
0.10	238	206	32	0.866	0.837	0.851	62/47	0.180
0.25	217	195	22	0.899	0.793	0.842	63/49	0.219
0.50	181	167	14	0.923	0.679	0.782	57/63	0.648
0.75	153	142	11	0.928	0.577	0.712	50/78	<i>0.017</i>
0.90	133	126	7	0.947	0.512	0.665	47/87	<i>0.001</i>
deployed filter (single point)				0.850	0.785	0.816	—	—

Table 3: The verifier’s operating curve on all 437 clean rows, pair-level labels, three checkpoints averaged. “fix/brk” counts items the verifier decides correctly and the filter does not, and the reverse; p is McNemar with continuity correction. **These p -values are a sweep, not eight tests we are entitled to read individually**—we report the column so the shape is visible, and rest no claim on any cell of it. The two italicised rows are significant *against* us: past $\tau = 0.75$ recall has collapsed far enough that the filter is the better decision rule.

struction.¹

Pair-contrast rows are built by asking the teacher separately about each candidate a pooled passage licenses. Where the labels differ, the passage enters as several rows identical in segment B and differing only in segment A and the target — rows a query-ignoring model cannot fit at all.

Distillation. The student is a 110M-parameter cross-encoder reading the candidate triple as segment A and the passage as segment B, roughly $290\times$ smaller than its teacher, trained with a split grouped on taxon pair so no pair spans training and development. Three seeds throughout.

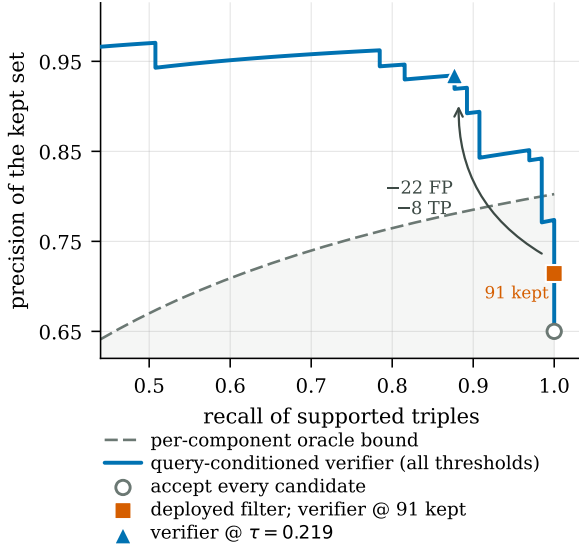


Figure 2: Precision against recall on BIOTX100 at the triple level (three checkpoints, probabilities averaged). The dashed curve is the per-component oracle bound $65r/(65r+16)$ of §4: a verifier with perfect knowledge of both taxa and the relation still admits all 16 pair-binding failures at every operating point. The deployed filter falls below it. The point of the figure is that the verifier is a curve where the filter is a point; the headline numbers in Table 3 are pair-level on all 437 rows.

6 Results

6.1 Precision side

Table 3 gives the operating curve on the full benchmark. The verifier spans precision 0.779 to 0.947 as τ moves, against the deployed filter’s single point at 0.850; that span, not any point on it, is the contribution.

For a single comparable number we fit τ on two of the three source blocks and apply it to the third, so that every row is scored by a threshold that never saw it and every row is used. That gives precision 0.825, recall 0.898 and F1 0.860 against the filter’s 0.816, with 62 items fixed and 47 broken. **The difference is not significant** (McNemar $p = 0.180$). We report it as the headline anyway, because it is the number the protocol produces; the honest reading is that on decision correctness the two systems are not distinguishable on 437 items, and a benchmark several times larger would be needed to separate them at this effect size.

Block by block, with the same held-out thresholds, the picture is not uniform: TEST299 0.787 $\rightarrow$

¹These correspond to repository tags V1, V2 and V3 respectively; the pair-contrast block was designed against an earlier, superseded error analysis, so its composition was not targeted at the decomposition reported in §4.

Query given to the verifier	P	R	F1
AUPRC			
Retrieved triple (deployed)	0.842	0.890	0.866
Generic relation, pair kept	0.876	0.687	0.770
Relation shuffled across items	0.879	0.679	0.766
Relation only, no taxa	0.917	0.045	0.085
Empty query (passage only)	0.653	0.191	0.296
	0.641		

Table 4: Ablating segment A at inference on all 437 rows, weights and threshold ($\tau = 0.04$) unchanged across every row. The base rate is 0.563. Deleting the taxa takes ranking from 0.912 to 0.677, and the empty-query arm—which is exactly the deployed filter’s formulation, passage in and verdict out—reaches 0.641, a tenth of the way from chance to the conditioned model. Per-seed standard deviations on AUPRC are ≤ 0.027 .

	AUPRC	F1 at $\tau = 0.5$
seed 1	0.9140	0.7832
seed 2	0.9030	0.8018
seed 3	0.9128	0.7755
mean $\pm$ sd	0.9099 ± 0.0060	0.7868 ± 0.0136
3-checkpoint ensemble	0.9123	0.7822

Table 5: Per-seed figures for VERIFIER-PC, and the ensemble formed by averaging the three checkpoints’ probabilities. Elsewhere in this paper the ensemble row is what is reported, and we name it as an ensemble rather than as a mean over seeds, because the two are different objects and an earlier version of this work conflated them.

0.859, REJECT50 0.000 $\rightarrow$ 0.700, and BIOTX100 0.915 $\rightarrow$ 0.884. The verifier is *worse* on the block whose distribution the deployed filter was in effect selected against, and better on the two it was not. We report the losing cell because omitting it would be selective.

6.2 The query, not the encoder

The deployed filter and the verifier differ in input format, architecture, training corpus and label source at once, so the headline is a deployment comparison and not a controlled ablation. The one factor we *can* isolate is the query, by removing it at inference with weights and threshold unchanged (Table 4). Deleting both taxa from the query collapses ranking from AUPRC 0.912 to 0.677 against a base rate of 0.563, and F1 from 0.866 to 0.085. The arm that matters most is the empty one: with segment A blank the model is doing exactly what

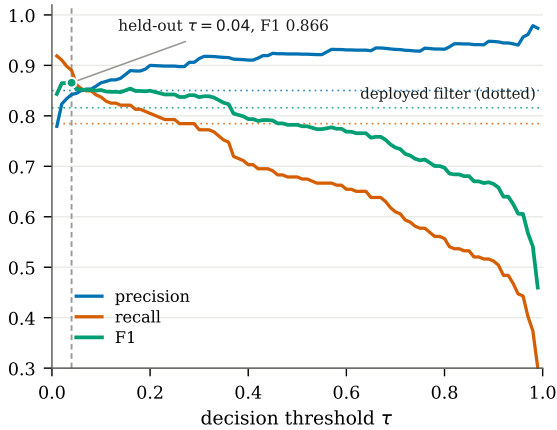


Figure 3: Precision, recall and F1 against the decision threshold on all 437 clean rows. The deployed filter is the three dotted lines: one point, not a curve. The marked threshold is fitted on the other two source blocks, never on the rows it scores. Precision is adjustable from 0.78 to 0.95; what a curator buys by moving right is precision paid for in recall, and the figure is the deliverable rather than any single row of Table 3.

the deployed filter does—reading a passage and returning a verdict—and it reaches AUPRC 0.641, roughly a tenth of the distance from the base rate to the conditioned model. The same weights, the same threshold; only the question changes. Substituting a generic relation for the retrieved one, pair held fixed, costs 0.096 F1, and shuffling the relation across items costs 0.100. The decision is carried overwhelmingly by the pair and materially by the relation.

The same evaluation does *not* support our training-data story. VERIFIER-BASE is query-conditioned as well—it differs from VERIFIER-PC in training corpus, not in input formulation—and on the full benchmark the two are indistinguishable: per-seed AUPRC 0.9137 ± 0.0012 against 0.9099 ± 0.0060 , block-held-out F1 0.857 against 0.860, and an identical 0.700 on the recall set. Pair-contrast supervision, which an earlier version of this work credited with the operating-point behaviour, buys nothing measurable here. What the reformulation buys is in Table 4; what the extra corpus buys is, on this benchmark, nothing we can demonstrate.

6.3 Recall side

REJECT50 asks the complementary question: of the candidates the pipeline threw away, can the verifier find the ones it should have kept? This is the one place the two systems are not close. The

deployed filter scores F1 0.000 by construction, having rejected all of them.

Nine of the 50 rows are also in the verifier’s training data, two of them among the true positives, so we report only the 41 clean rows: 9 true positives wrongly discarded and 32 correct rejections. At the held-out $\tau = 0.02$ the verifier recovers 7 of the 9 while wrongly re-admitting 4 of the 32, for F1 0.700 where the filter has none.

We do not want this read as a precise rate. Nine positives is a small numerator, and the Clopper–Pearson interval on 7/9 is $[0.40, 0.97]$. What it establishes is qualitative and, for a curator, the most useful thing in the paper: a class of true positives that the deployed filter cannot return *at any threshold*, because it never assigns them a score at all, becomes reachable once the decision is conditioned on the triple.

6.4 The operating point, against our own interest

At 91 accepts the verifier’s confusion matrix is 65/26/0/9 — identical to the deployed filter’s. This is arithmetic rather than agreement: with 65 supported triples among 100, any system accepting 91 candidates incurs at least $91 - 65 = 26$ false positives, and both systems attain that floor. The two accept sets are nevertheless not the same, agreeing on 90 of 100 items. We therefore do not beat the deployed filter at its own operating point, and no system could.

What changes is that its operating point is fixed and ours is not. Three consequences are available to a curator and none were before. At matched *recall* of 1.000 the verifier needs 260 reviews at precision 0.842 rather than 213 at 0.850: comparable precision at much higher recall cost — though at $n = 100$ this difference is itself not significant ($p = 0.12$). Accepting a 17% recall loss gives 181 reviews at precision 0.923, a third less work at half the error rate. And a curator who wants everything can still have the 87-candidate point at 0.747. The deliverable is a calibrated score with an adjustable threshold, not a better verdict, and a curator-facing filter should expose the former.

The right threshold is not a property of the model: ingestion into a reference database pays for false positives and belongs near $\tau = 0.85$, while assembling candidates for expert review pays for false negatives and belongs near $\tau = 0.001$, where 87 candidates survive at precision 0.747 and nothing supported is lost. Those points differ by 33 candi-

dates and 0.20 precision. A filter emitting a verdict fixes that choice at training time, for users who were never asked.

7 Where the reformulation stops

Four false positives survive the reformulation: one pair-binding, one predicate, two entity (Figure 1). It is tempting to read the per-class repair rates — 10/11, 7/8, 5/7 — as showing which class the reformulation fixes, and we decline: every pairwise comparison is null (Fisher’s exact $p = 1.00, 0.53, 0.57$), as cell sizes of 11, 8 and 7 make inevitable. The decomposition’s value is the *partition*, not a differential — 16 of the 35 unsupported candidates have every component correct, so no component-wise check reaches them at any operating point — and it is that structural fact the reformulation is designed against.

Entity disambiguation, and a gate that cannot fix it. The natural remedy for the entity class is to gate candidates on taxonomic validity. We tested it against the Open Tree Taxonomy (Rees and Cranston, 2017). It catches 0 of the 7 entity failures and 1 of the 35 unsupported candidates overall, while discarding 5 of the 65 supported ones. The reason is diagnostic rather than incidental: *Illiger*, *Dikarya*, *copia* and *sulla* — the surnames, clades, transposon families and vernacular names that produced the entity failures — all *resolve* in the taxonomy. The failure is disambiguation, of the kind species-name normalisers are built for (Gerner et al., 2010), not validity. The gate’s casualties are meanwhile viruses (SARS-CoV-2, tomato spotted wilt virus, orchid fleck virus, Colorado tick fever virus), which are legitimate interaction partners in this domain.

Argument order: an open problem, and why. An ordered query in principle yields direction by argmax over the two orderings. We report no such number, because the gold standard cannot currently score it. Expert curation marks reversed triples as correct — we verified two by hand, (*Breinlia*, host of, *Rattus*) and (*Demidospermus*, parasitized by, *A. inermis*) — and an LLM adjudicator run over adjudicable items contradicts plain surface order in 19 of 64 cases (29.7%, [18.9%, 42.4%]).

The difficulty is systematic rather than random, and this is the part worth reporting. Among the 5,591 teacher-labelled reversals, the rate at which the teacher rejects a reversal is governed by the

relation’s *surface string*, not by the relation. Restricted to surface forms with $n \geq 30$, direction-encoding participles and prepositional forms are rejected at 0.51–0.70 (*affecting* 0.70, *affected by* 0.61, *parasite of* 0.57, *feeds on* 0.57), while bare relational nouns are rejected at 0.04–0.10 (*host for* 0.05, *infestation* 0.07, *host of* 0.08, *pollinators* 0.04). Whether the supervision encodes argument order at all depends on which string the rule layer happened to match. A 193-item human direction annotation is in progress; until it exists, a directional score on this data would measure the annotation convention rather than the model. **We make no direction claim in this paper.**

8 Related work

Filters that answer an easier question. That a classifier can score well by exploiting how its data were assembled is among the best-established results in NLP. Partial-input baselines recover much of the label from the hypothesis alone in natural language inference (Gururangan et al., 2018; Poliak et al., 2018) and from the passage alone in reading comprehension (Kaushik and Lipton, 2018), and such rules hold in distribution while failing under deployment (Geirhos et al., 2020; Du et al., 2024); the closest task-level precedent is fact verification, where a claim-only classifier matches evidence-aware models because of how the corpus was built (Schuster et al., 2019). Section 3 reports that phenomenon rather than a new one. Two things are specific here: the missing input is the query the pipeline itself issued rather than an annotator artefact (Wiegand et al., 2019; Parmar et al., 2023), and the cue the filter does have is inert by construction, since the rule layer fires on every candidate in the pool. A cue present on every input can certify nothing — though Feng et al. (2019) caution that partial-input results license claims about the model and the corpus rather than about the task, and we keep ours to the former.

The argument-ignoring failure is already named. Distant supervision assumes that a sentence containing two related entities expresses the relation (Mintz et al., 2009), and its standard relaxation exists precisely because it does not (Riedel et al., 2010). Since TACRED an instance has been the triple (sentence, e_1 , e_2) rather than the sentence (Zhang et al., 2017), and how the pair is injected into the encoder is worth several F1 points (Baldini Soares et al., 2019; Wu and He, 2019; Zhou

and Chen, 2022). Rosenman et al. (2020) name our failure exactly, as an “event heuristic” that classifies on (sentence, relation) while ignoring the arguments, and show that QA-style models repair it on a challenge set. *We claim no part of that diagnosis.* What is ours is that the heuristic is observable in a deployed system’s own expert-graded output rather than in an adversarially constructed set; that the failures partition into classes with a quantified bound on any component-wise remedy (§4); and that we report the operating-point analysis honestly (§6.4). The entity-type shortcut usually invoked to explain such errors (Peng et al., 2020; Mtumbuka and Schockaert, 2024) is unavailable here, because both slots are filled from one inventory of taxon surface forms and type separates nothing.

Attributing failures across pipeline stages is standard, and the accounting is easy to get wrong (Taillé et al., 2020; Ai and Kavuluru, 2023), but it is done across stages on benchmark data; we attribute errors within a production filter’s false positives, against expert grades of the same items, and report against our own interest that only 11 of 50 curated rejections are ours to fix (§6.3).

Conditioning on the query, and what to hand a curator. Casting a relation as a question about a named entity (Levy et al., 2017), or as an entailment hypothesis verbalising the relation between two of them (Obamuyide and Vlachos, 2018; Sainz et al., 2021), is the reformulation our input format instantiates, and the resulting decision — does this passage support this specific claim — has the shape of scientific claim verification (Wadden et al., 2020). Our verifier is a supervised cross-encoder reading the triple as segment A (Nogueira and Cho, 2019), trained on labels from a prompted teacher, as is established practice for distilling a large model’s judgments into a small student (West et al., 2022; Gekhman et al., 2023). Selective classification trades coverage against risk by thresholding a score instead of issuing a verdict (Geifman and El-Yaniv, 2017), and that is how automated extraction is made useful to human abstractors: Swaminathan et al. (2024) abstain under calibrated thresholds to automate 57–95% of clinical chart abstraction at controlled error. The filter we audit offers no such control. What the reformulation buys is not a better point but a curve.

Text mining for interaction databases. GloBI (Poelen et al., 2014) is populated in part by text mining (Dimitrova et al., 2020); the system au-

ditioned here (Ruch et al., 2024) retrieves over a biodiversity-indexed MEDLINE (Gobeill et al., 2020). Judging an extracted fact before it reaches a curator is an old idea (Rodriguez-Esteban et al., 2006), as is evaluating such tools by whether they help the curator (Hirschman et al., 2012). The closest precedent for our design is Britan et al. (2018), a curation-support tool scored by the curators of the database it serves; Lee et al. (2018) likewise evaluate triage against a resource’s existing practice. Cuzick et al. (2023), independently, take the interacting pair rather than the passage as the annotation unit — the unit our pair-binding class isolates. Large language models are applied to interaction extraction at scale (Keck et al., 2025; Farrell et al., 2024; D’Souza et al., 2025; Zou et al., 2026), but as extractors: Zou et al. (2026) standardise names and correct categories post hoc without checking extracted interactions against the text that produced them. We are not aware of published work that verifies literature-mined biotic-interaction triples against their source passage before ingestion.

9 Conclusion

A deployed filter accepts 91 of 100 candidates and reaches a precision this benchmark cannot separate from accepting all of them, because the question it is asked is guaranteed true by the rule layer above it. Its false positives partition into three mechanisms, 16 of the 35 unsupported candidates being invisible to any component-wise check at any operating point. Conditioning the decision on the candidate triple repairs 22 of the 26 and recovers triples the pipeline discarded. At the deployed operating point the two are arithmetically identical: what changes is that a curator has a dial where there was a verdict.

Limitations

Both benchmarks are small and singly annotated. The benchmark is 437 rows in three blocks, every gold label assigned by one domain expert with no item graded twice, so we report no inter-annotator agreement and cannot bound label noise. The headline comparison is *not* significant — McNemar against the deployed filter gives $p = 0.180$ — and we present it as a tie on decision correctness rather than as a win. Separating two systems at this effect size would need a benchmark several times larger. On the recall side the denominator is 9 positives and the interval on 7/9 is [0.40, 0.97].

The failure classes are derived, not annotated.

Mechanism is assigned by a fixed cascade over the expert’s three component grades rather than by direct annotation, so “pair-binding” is a residual class and any component-grading error accumulates in it. That class is also the one that fixes the 0.802 bound. Cell sizes are 11, 8 and 7, and we claim no differential effectiveness across them (§7). The per-row mechanism grades behind Figure 1 were lost before release; the figure is reproducible only from the aggregate counts printed in its caption, which is a reproducibility defect we prefer to state than to paper over.

The deployment comparison is not a controlled ablation.

The deployed filter and the verifier differ in input formulation, architecture, training corpus and label source at once. We can show that the query is necessary *within* the verifier (Table 4) and that the filter’s own feature is inert on the deployment pool, but we cannot decompose the improvement across those four factors. The missing cell is a sentence-only student trained on the same teacher labels. Table 4’s empty-query arm is the closest thing we have — the same weights asked the sentence-level question, reaching AUPRC 0.641 — but it is an inference-time manipulation of a model trained with the query, not a model trained without one, and it cannot substitute for the trained control.

The training labels are wrong about one time in seven.

Our training data are labelled by a prompted teacher. Against 150 human-gold items its mislabel rate is 14.7%, [9.4%, 21.4%], split 10 false positives and 12 false negatives, and the student inherits that boundary including its systematic errors; training encoders on model-generated labels can degrade and destabilise them (Lu and Smith, 2025). The benchmark is graded by a domain expert rather than by the teacher, so the reported precision is measured after whatever damage the noise does — but the verifier’s ceiling is the teacher’s judgment rather than a curator’s.

Leakage on the recall-side benchmark. Nine of REJECT50’s 50 passages appear in the verifier’s training data, including 2 of its 11 true positives. We report the leakage-free figures in §6.3 and recommend the leaked subset for any future comparison. BIOTX100 is *not* unaffected: three of its 100 passages are exact duplicates of training text at 5-gram Jaccard 1.000, found by scanning the benchmark against all 48,338 training passages

rather than by the shipped `in_train` flag, which misses them. Those three are excluded throughout. An earlier version of this paper reported 0/100.

The evaluation is pair-level; the diagnosis is triple-level.

Only BIOTX100 carries gold at the level of the whole triple. TEST299 was curated as a pair-level set and the relation term stored on its rows was attached by the retrieval pipeline, not by an annotator, so scoring it against that term would test the model on a label nobody checked. We therefore evaluate at the pair level everywhere and diagnose at the triple level only in §4. The cost is real: 11 of the deployed filter’s 26 triple-level errors are correct pair-level positives, so the headline understates what triple conditioning buys. The right fix is triple-level gold on the other two blocks, which we do not have.

One earlier set of numbers is withdrawn.

An earlier version of this work fixed the decision threshold by applying a prior-shift rule to the reporting set’s own gold labels, and scored BIOTX100 against the full-triple column while describing the benchmark as species-level. Every number computed that way is withdrawn, including a reported McNemar $p = 0.018$ that does not survive either correction. The present numbers share no threshold and no label column with it.

The classifier owns a minority of the pipeline’s failures.

Of 50 curated rejections, 11 are classifier errors. The rest fail upstream: 32 carry an incorrect interaction term and 17 a mis-resolved taxon, 14 of them both, and a further 4 are correct rejections with every component individually right. No classifier change, ours included, addresses the upstream mass, and our precision gain is not a pipeline-level gain. An earlier internal study using a different assignment convention reported a 40/34/22 split across the rule, entity and classifier layers; the overlap-explicit counts above are what we can reproduce, and only the 22% classifier share is stable across conventions.

Argument order is unevaluable on the current gold standard.

Expert curation scores reversed triples as correct, so no directional number in this paper would measure the model rather than the annotation convention (§7).

Prior numbers from this project are void.

Earlier results were computed with canonical taxon and relation strings that are absent from the pas-

sage on up to 54 of 100 rows. Every number here uses the surface-form query; comparisons to our own prior reports should not be made.

One system, one domain, one pretrained teacher.

We audit one retrieval system, one interaction vocabulary and one domain, taxon–taxon interactions in MEDLINE, so we cannot show that filters over other pools fail the same way, although the mechanism — a pair-agnostic decision over passages naming more than one candidate pair — is not specific to ours. The passages are published MEDLINE text and the teacher is a pretrained model; we cannot exclude that it saw them, or the interactions they report, during pretraining, so its judgments should be read as an optimistic estimate of performance on genuinely new literature.

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A Negative results and cost

Table 6 collects four interventions that did not pay, reported because each is a plausible first thing to try.

Intervention	Benefit	Cost
Taxon-validity gate	0/7 entity failures caught	−5/65 supported triples; rejects viruses
Self-distillation	none	−0.001 P, −0.008 F1, −0.007 AUPRC
6-layer compression	1.9× CPU (24.2 → 46.2 pairs/s)	−0.088 precision
Corpus signal-matching	none	−0.059 AUPRC, both architectures

Table 6: Four plausible design choices that measurably do not pay.

These deltas were measured under the earlier evaluation described in Limitations, so we give them as directions rather than exact magnitudes. The validity gate is treated in §7. Self-distillation at equal capacity is null, suggesting the teacher’s knowledge is already saturated in the student, which bounds what a same-architecture approach can add. Compression to six layers buys 1.9× CPU throughput for 0.088 precision; since the uncompressed verifier already runs at 24.2 pairs per second on CPU, above this pipeline’s ingestion rate, the trade is available but unattractive. Reweighting the corpus toward the deployment signal distribution cost 0.059 AUPRC on both architectures we tried.